Coordinates and residues in the Higman–Sims catalogue

Private L091 component

Let $G = \mathrm{HS}$. We use the supplement dated 6 October 2003, with 23 pages, as the fixed source for the diagram numbers below. Its statements and notation remain attributed to the original authors. The complete classification, the local numerical data, the bare isomorphism distinctions, and the truncation certificates are acquired from their separately reviewed components. This application identifies the source coordinates and verifies the small named residues. It does not replace those earlier proofs.

Theorem 1. *The numerical profiles of the 86 diagrams of ranks two through six, with the six explicit amendments below, give a bijection to the 86 classes in the acquired classification. The thirteen printed truncation assertions have actual incidence isomorphisms. The thirteen occurrences of a Petersen residue and the two occurrences of an affine-plane residue have explicit isomorphisms to the Petersen incidence structure and* $\mathrm{AG}(2, 4)$*, respectively.*

Coordinates and the two ambiguities. For each source diagram, first compare the multiset of type sizes with every acquired class. Apart from two pairs, this identifies a unique class. The entry 44320 in diagram 2.6 is retained in the transcription but replaced by 44352 for this comparison: the printed integer does not divide $|G| = 44352000$, so it cannot be an index for the asserted transitive action.

The two rank-three diagrams 3.22 and 3.23 have identical type sizes and local numerical data. Their candidate classes are 14 and 15. The acquired full automorphism proof gives $\mathrm{Aut}(\Gamma_{14}) = G$ and $\mathrm{Aut}(\Gamma_{15}) = \mathrm{Aut}(G)$, of respective orders 44352000 and 88704000. These are full typed groups, rather than merely normalizers of a specified action. Thus the source's two different automorphism assertions select 14 for 3.22 and 15 for 3.23. For the remaining tied pair, 5.2 and 5.3, the complete diagram has respectively one and two quadrangle bonds. Every permutation among equal-sized types is checked. This selects classes 83 and 84.

The finite coordinate list records a permutation from each source's ordered nodes to the actual model's types. The resulting 86 class indices are precisely $1, \ldots, 86$, each once. This identifies classes by proved invariants within a complete classification; it makes no claim that an unavailable historical generator list is literally the same list of permutations as ours.

Literal discrepancies. An ordered triple (a, g, b) records the eccentricities of the two specified vertex types and half the girth of their incidence graph. The finite record retains both the printed and the actual values. The six amendments are as follows.

Diagram	Amendment
2.5	The node order at type 36960 is 15, in place of 11.
2.6	The second type has size 44352, in place of 44320.
3.15	The right arm, ordered from the type of size 352 to that of size 1100, has triple $(5, 4, 6)$, in place of $(6, 4, 5)$.
3.30	The top node order is 3, in place of 1.
4.22	The bottom bond is a digon; the omitted diagonal joining the two types of size 352 has a single bond.
4.28	The bond from type 352 to type 1386000 has triple $(7, 5, 8)$; the bond from type 1386000 to type 1100 has triple $(4, 3, 3)$ and is a complete-graph incidence structure.

Why these are amendments. For each of these six rows the type-size multiset, after the explicitly declared index amendment in 2.6, selects a unique class. All compatible type permutations are tested against the acquired local data. Therefore the stated mismatch cannot be removed by choosing another class or by silently exchanging equal-sized types. In particular, both possible exchanges of the 352 types

in 4.22 remain inconsistent with its printed bonds. The amended values are consequences of actual residue graphs, not an inference from visual similarity to another diagram.

The independent source audit also records order inconsistencies in the displayed parabolic, Borel or kernel names of 3.1, 3.4, 3.5, 3.12, 3.13, 3.15, 3.16, 3.20, 4.24 and 4.28. We retain those discrepancies. This theorem does not supply the abstract names that should replace them, nor prove the remaining names merely because their orders agree. The source's boxed kernel and generic extension dot keep their own meaning; the dot alone is not treated as an assertion of nonsplitting. The Aut and Cor values left blank in 3.10 are still recorded as blank.

Truncations. Restricting a coset geometry to a subset J of its types retains the same cosets and the same incidence relation. Each previously reviewed positive truncation certificate gives an automorphism α of G and a type permutation carrying the retained parabolics to those of the target. The maps

$$G_i x \longmapsto \alpha(G_i)\alpha(x)$$

are well defined bijections. Two cosets meet if and only if their images meet, so these are incidence isomorphisms. Composing the already verified certificates with the coordinate list above proves all thirteen printed links. The record gives the exact retained source nodes for each link. No new transporter search is needed. The eight printed doubling assertions are separate obligations.

Petersen and affine residues. Take as Petersen vertices the ten two-element subsets of a five-element set, with adjacency defined by disjointness. The incidence structure has these ten vertices as points and its fifteen graph edges as lines. This is the standard Petersen model used in the thirteen certificates.

For the affine model, set $F = \mathbb{F}_2[t]/(t^2 + t + 1)$. The polynomial has no root in $\mathbb{F}_2$, so F is a field of order four. The points are F^2; its twenty lines are $y = ax + b$ and $x = b$. Two distinct points lie on a unique such line, by division of their coordinate difference when the first coordinates differ, and by the vertical line otherwise. Thus this is $\mathrm{AG}(2, 4)$.

For each of the fifteen source occurrences the certificate supplies a bijection from the full displayed model to the actual residue graph. For every vertex, the image of its entire neighbor set equals the entire neighbor set of its image. This proves incidence preservation in both directions. The model points are also checked to occupy one entire actual type. Consequently the record gives the exact point polarity, while retaining every printed star; no unstated convention about reading stars on sloping diagram bonds is imposed.

Evidence and scope. The source transcription was checked on all 23 native pages by two readers. The new coordinate calculation uses the accepted complete finite records. Graph software only proposed the fifteen small positive maps; their verification uses literal adjacency lists and the explicit models above. Damaged map, roster and coordinate controls accompany ordinary and optimized executions. The finite inventory and each acquired review are fixed by exact file hashes. This component contributes no new census, whole-original completion, conjecture resolution, formalization or compression measurement. The source transcription, earlier classification and earlier local-data proofs retain their separate authorship and review records.